

Claude Code 29: Can Claude Code Find Facts? And If So, Should I Believe Them?

Who is the research assistant with 3D printed papers -- the human or Claude Code?



SCOTT CUNNINGHAM

MAR 05, 2026

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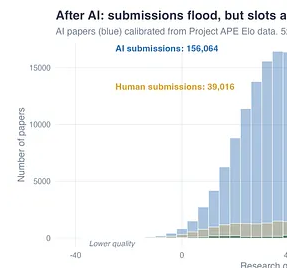
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Monday's post was what I called Claude Code fan fiction — a supply-and-demand analysis of what happens to academic publishing when the cost of producing a manuscript collapses. It went viral, which I did not expect. This post is the companion piece, but it's about something different. Something that's been bothering me more.

Someone on LinkedIn asked me to demonstrate a “purely automated paper” because they were skeptical. So I filmed it. Here's the video. It's long even with me pausing, but you'll get the gist if you watch the first part and skip around.

Claude Code 27: Research and Publishing Are Now Two Different Things

SCOTT CUNNINGHAM • MAR 2



Claude Code has made it easier to do research now. But it is about to get much harder to publish in traditionally valued locations.

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I went to bed and woke up with a paper

Monday night, 9:21 PM. I created a directory, pointed Claude Code to it, and gave it the vaguest prompt I could come up with. I told it I study labor and mental health, that I'm interested in drug policy, that I like Autor and Acemoglu and Ricardo and Smith. I said I wanted to see if it could make a paper without me putting hands on the wheel. I gave it access to Mixtape Sessions repos so it could see good code and real slides. And then I put the Stephen King limited series, **11.22.63**, on and waited.

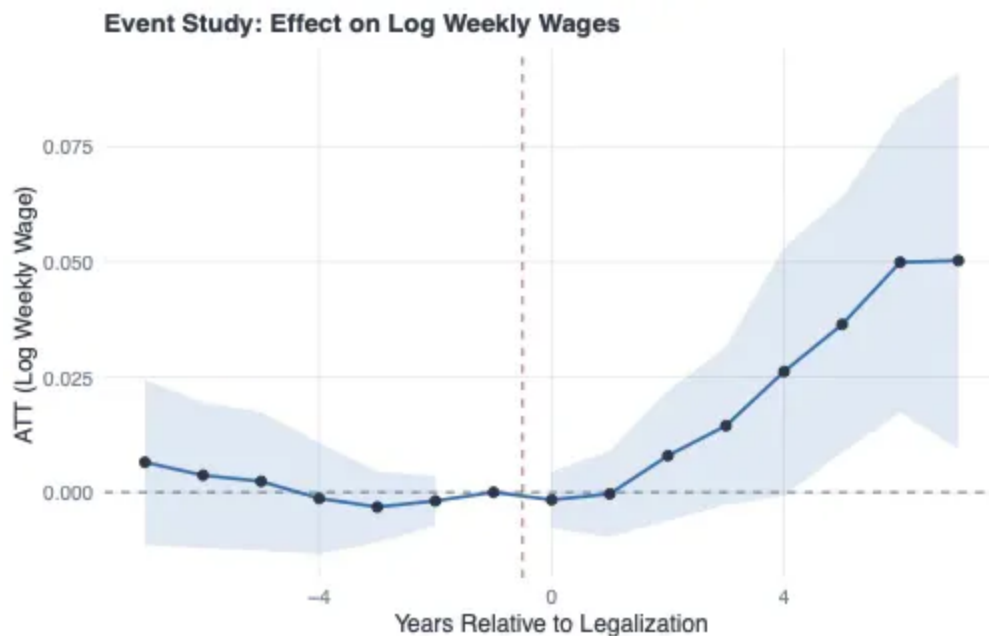
Claude Code chose a topic — marijuana legalization's effect on employment and mortality. It crawled the web and found the data. It came up with a research design: diff-in-diff using Callaway and Sant'Anna with robustness from TWFE, Sun-Abraham, and Gardner. It was more or less done by 10:45, but I went to bed at 10:30.

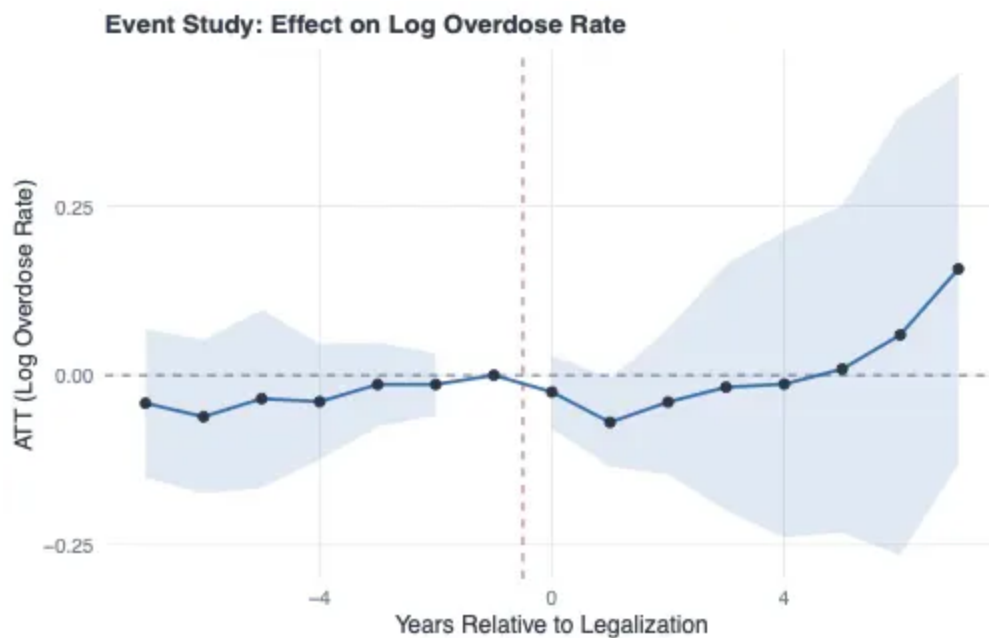
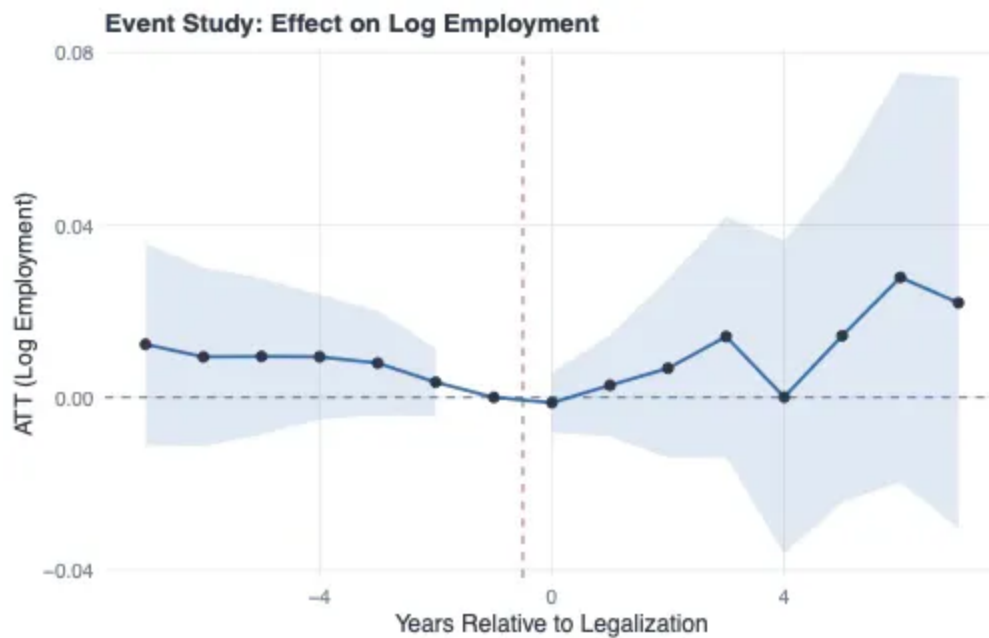
It's true that I did not completely take my hands off the wheel. I was pretty regularly saying things. But it wasn't prompting — not exactly. It wasn't writing either. It was

automation minus epsilon. I told it to write like my favorite economics writer, [Martin Weitzman](#). He wrote with this difficult to describe kind of rhetoric — moral urgency, seriousness, perfect prose, as if what he was talking about right then was the most important thing that could be talked about, because of how much was at stake, while still managing to sound like a scientist. A very weird skill that he pulled off his entire career from day one until the end, and that was the way I wanted this paper to sound.

When I woke up, I ran it through my Referee 2 persona and submitted it to Refine.ink. That process, including my checking, caught real bugs — one being a Sun-Abraham aggregation error that doubled the treatment effect, 295 false zeros in the incarceration data from a coverage boundary. I paid about \$100 total across two rounds of [Refine.ink](#), had Claude fix everything. Done. Total active time: about 3.5 hours. You can watch the video to see me do most of it.

I haven't carefully read the [paper](#) yet. But I've seen the event study plots. And the one on wages has been hard to ignore and made me wonder — *will I read this paper? Will I work on this paper now? Should I?*





Are we asking the wrong question?

[Project APE at the University of Zurich](#) is pitting fully automated papers against AER and AEJ articles in head-to-head matchups. The AI papers win 4.7% of the time, but over time that's grown to over 7%. Though the Elo gap is massive, occasionally a paper pops up in the right tail — as you'd expect from the normal distribution which has infinite tails. And so naturally, the conversation has become: can AI papers compete at the top 5? And that is also naturally met with incredulity.

But I don't know if I agree that that is the right question. I don't think that the policy relevant or the scientifically relevant question is whether 3D printed manuscripts will ever be published in the AER since most economists never publish in the AER. It is not the modal experience for research productive economists to publish in top 5s.

The majority of research economists do not publish in the AER. The mode number of AER publications over a lifetime career is, I would guess, zero. People have very productive careers publishing in the second tier top 10 journals like REStat and EJ, but even then the backbone of most economists' vitas is spent publishing at the field journals — JHR, JOLE, JPubE, Economic Inquiry, Southern, Labour Economics, Health Economics, etc. The top 40, not the top 5. Tenure decisions at very good schools are rarely requiring their junior faculty to have a top 5.

So the real question isn't whether an automated paper can beat an AER article in a blind matchup. It's whether it can compete at the *Journal of Human Resources*. Can it get past the desk at *Economic Inquiry*? At JOLE? And when I look at this [marijuana paper](#) — standard diff-in-diff, publicly available data, clean pre-trends, a question that fits squarely into a well-established literature — the answer is: *possibly*. I don't know yet. Marijuana policy isn't my field, and I haven't read what the paper says carefully enough to know how it compares to the existing literature. But the methods are boilerplate. The data is real. The event studies look reasonable. And I audited the code with sub-agents three times, replicating the code in multiple programming languages, just to find the bugs. There doesn't appear to be any bugs. The data is real.

Now, was the data correctly handled? Were the laws coded correctly? Claude used data from the Harvard Dataverse cannabis policy for this project, but did it process it correctly? How well did it clean the data? Are these wages correct? On and on. Those I don't know, but the thing is, those are also trivially answerable. They're easier to answer now than they ever have been, and if they weren't handled well, then they can be.

But the thing is — this *kind* of paper has a literature. It's an active research agenda for several people. RAND works on this topic. The repercussions of drug legalization is an important policy question. It matters for people's lives and the organization of society itself. And the stakes are high getting the answer wrong. So again I ask — could this paper make it past the desk, blinded, at JOLE? Maybe, maybe not. What about Journal

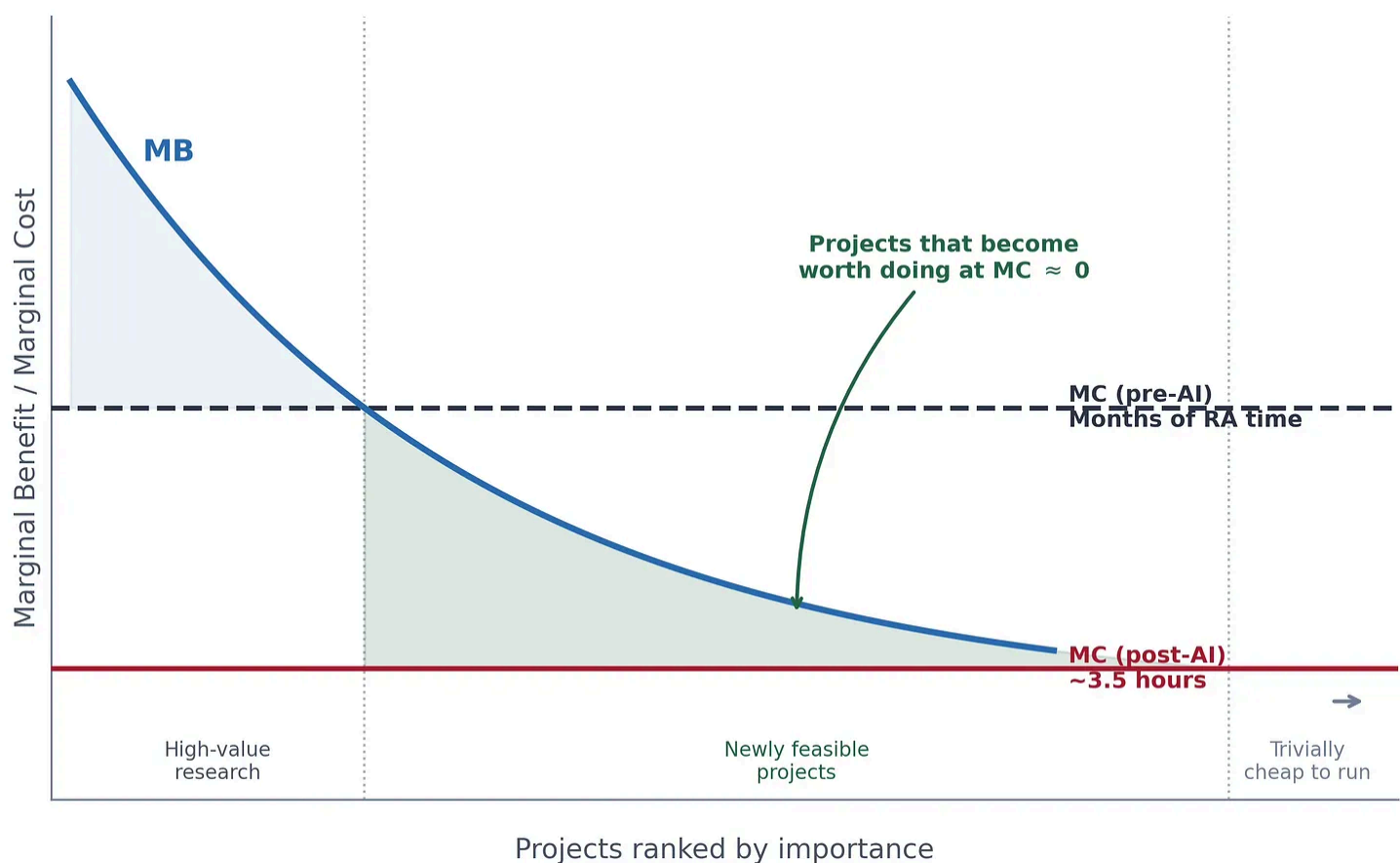
of Population Economics? What about Journal of Health Economics? Consider these papers.

1. **Mathur & Ruhm (2023), “Marijuana legalization and opioid deaths,” *Journal of Health Economics*** — This is the closest hit. They find that medical marijuana with retail dispensaries is associated with *higher* opioid mortality, not lower. Directly relevant to the overdose outcome in the 3D printed paper that Claude wrote and recent.
2. **Chakraborty, Doremus & Stith (2023), “The effects of recreational marijuana legalization on employment and earnings”** — This started as NBER WP 30813 and was published in *Journal of Population Economics* (2025). They find little adverse effect on labor markets for most working-age adults, with some agricultural employment gains. Very similar question to Claude’s 3D printed paper.
3. **Anderson, Hansen & Rees (2022), “The public health effects of legalizing marijuana,” *Journal of Economic Literature*** — A comprehensive survey covering crime, mortality, and labor market effects.

So that is the real question to me — could these 3D printed papers make it past the desk, to referees, get R&Rs, and get published at top field journals? We don’t know the answer to that question, and it’s not obvious yet what parts of that question are important, and what parts of that question are not important.

What happens when marginal cost is zero?

But put journals aside. I want to say something else. What interests me more is something simpler. If the marginal cost of producing a submission-quality manuscript — complete with identification strategy, real data, replicable code — has collapsed to zero, then we should expect this to be used for any task where the marginal benefit is just above zero. Look at this graphic and think about it for a second.



Historically, any kind of econometric application would only be undertaken for high-value things. Why? Well, it took so long, could be screwed up in so many ways, and the refereeing process chewed up years of your life even ignoring the time it took to get to a manuscript. Even something as straightforward as diff-in-diff only really made sense to undertake if the question was important enough to justify months of data collection, cleaning, coding and writing.

But I did this in 3.5 hours while watching Netflix. You can drum up a diff-in-diff with actual data, done more or less correctly with robustness checks and event studies, while you sleep. Which means the marginal cost of producing a manuscript is now zero. Which means we should see these things used for trivial things — low value projects, way down the demand curve, now probably are candidates for econometric analysis, no matter how trivial they may be.

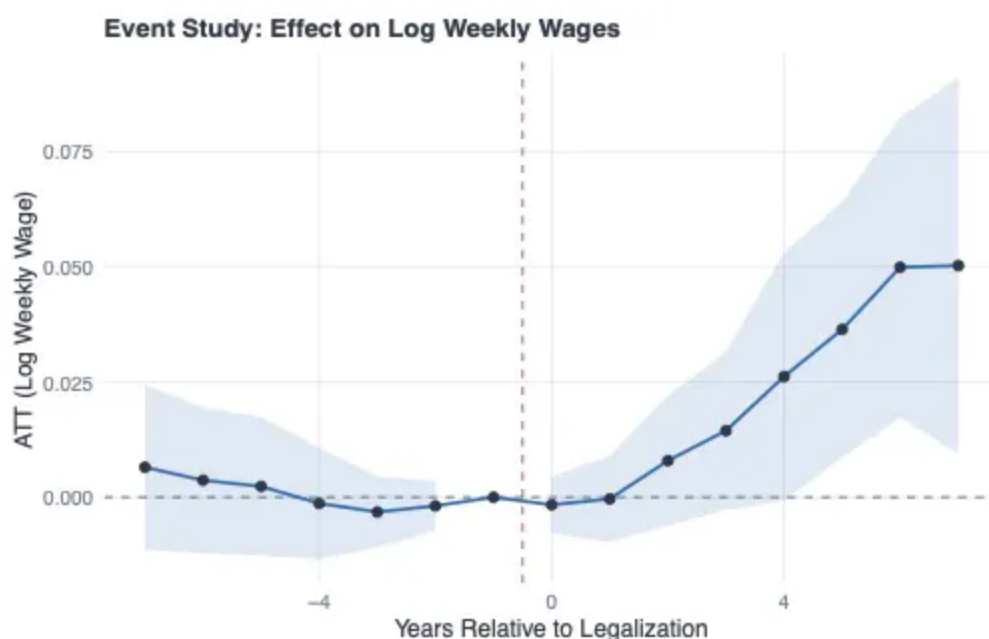
So if that's the case, then we should see causal inference move into tasks where it can be useful but where the marginal value was always sort of small. Not career-making research. Not AER papers. Maybe things that keep you from getting fired from your consulting job. Things that settle an argument in a policy memo. Things that answer a question a city council member asked and nobody ever bothered to study because the

effort-to-importance ratio was too high. Diff-in-diff is the most popular quasi-experimental method. It is easy to do and it's relevant in a lot of situations. If the marginal cost of doing it is zero, then aren't we heading to the same place that computing simple averages and making line graphs went — something you just do, because the data is there and the cost of not doing it is the same as doing it. Couldn't diff-in-diff and instrumental variables just become household names if the marginal cost of doing them is zero? I mean why wouldn't I run a synthetic control when deciding on whether or not to have pepperoni pizza at my kids' little league party? Why not present shift-share results at the PTA meeting? I mean those are low value uses, and now the cost of doing them is zero, so ... won't they happen?

We built an entire graduate curriculum around methods that took months to implement properly. The scarcity of execution time was itself a screening device — only questions deemed important enough got studied. That filter is gone. What replaces it?

Is the finding true?

Here's where it gets strange for me. I now have a *result*. The result is this: cannabis legalization *appears* to increase weekly wages by about 2.2 percent. Employment effects are null. Overdose mortality shows no discernible change. I have this graphic.



The data is not hallucinated. It's from the BLS and CDC WONDER. The methodology is boilerplate diff-in-diff with robustness across four estimators with the prominent one being Callaway and Sant'Anna. The event studies, particularly for wages, show clean pre-trends. It's not dissimilar from papers I've seen published in this literature.

But is this result *true*? Should I *believe* it? Should I *read* the manuscript it wrote? Should I now *work* on this project when I never came up with it, and if I do work on it, what am I obligated to say and do? Do I coauthor it with Claude? Am I first author? Am I Claude's research assistant, not the other way around?

It is one thing to do these on a substack, sort of showcasing what can be done, but it is quite another thing to drop everything and barrel head first into doing this, letting Claude Code shake you around so that you are focusing now on the project it came up with. Now it really is the blind leading the blind. And the fact that this can be drummed up so easily, with no effort — what do I make of “findings” now?

Al Roth once said empirical work exists for three reasons: 1) to establish facts about the world, 2) to tell theorists their models may need to be tweaked or abandoned, and 3) to whisper in the ears of princes. So here I am with a result produced by a machine. Is that event study plot a *fact*? Or does it only become a fact if it's published in a peer reviewed journal? Is a mean a fact then when it's not published in a journal? If I pull down my bank statement, and produce a piechart of spending, is that a fact even if it isn't published? Just what is knowledge now if it's produced with no effort at all and yet I don't think to send it out? Do I even need to send it out? Should I take this event study plot and tell a theorist their model is wrong? Should I show this event study to a state legislator considering legalization? Or do you only show event study plots to state legislators when the study is published — in which case what then we do about the unpublished NBER working papers that newspapers treat as published all the time?

At what point does the conversation shift from policing who wrote the paper to asking whether the finding is true? And what makes it true? Is it true because it's true or is it true because someone verified it and told me it was true? Do I need to read this paper? Will I read this paper? Will I burn this paper? Will I vow *never* to work on this topic because Claude wrote this paper?

Because if it is true — if cannabis legalization really does increase wages through labor market formalization and criminal record removal — then does it matter that no human designed the study? The data is real. The methods are credible. The result either holds or it doesn't.

But how do I know it's true? Do I need the peer review to say it is a fact? Did I need the peer review before it was published to believe things? It's all very strange and frustrating in part because there very well could be no matter how we slice it a massive avalanche of "purported facts" coming, whether to the journals or whether to our lives, in such volume that lacking a verification system, how will we sift through it? We will have more submissions than slots. It could create major bottlenecks in an already very long publication process for all of us. Is the time to publication going to go up now? I write a manuscript and now I'm standing in a very long line behind 10,000 robots?

But I caught the bug because I've spent twenty years learning this

I did catch a Sun-Abraham aggregation error, though. You can see it in the video. Something smelled off because it was so much different than the others, and it should have been nearly identical to CS. I knew to drop the Vera incarceration data when I saw the coverage gaps. I was still involved.

But the thing is — I only know to do that because I've been exerting myself for years on learning causal inference methodologies, particularly diff-in-diff. I've written a book about it and its sequel. I've published and taught until I was blue in the face. So I see things.

But there's another diff-in-diff coming. Will I see it then?

And another after that. And another. The free-rider problem cuts both ways here. If the machine does the work, what incentive do I have to keep investing in the human capital that lets me catch the bugs? My skills depreciate. Or worse — they never get created in the first place for the next generation. At what point does the work below the marginal cost curve just happen, regardless of whether anyone is qualified to verify it?

And then there is the paper the machine made for me. Should I read it? Should I *work* on it — seek to improve it, explore it more, push it further? Do I touch this? Does it make me a bad person if I do? These are questions I don't quite know what to do with because I've never asked them before, never had to think about them before.

The speed is the thing. Even if you can say “I don't think these are making AERs, so I'm not going to worry about it” — that doesn't address the fact that they may already be making papers suitable for JHR or JOLE or the Journal of Law and Economics. They might be. And that changes fast. And nobody has done a clean test. And the people offering their expert opinions have things to lose.

The potential is there. But so is the free riding. And so are the questions I don't have answers to.

I suspect we all find out soon.



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"I told it to write like my favorite economics writer, Martin Weitzman. He wrote with this difficult to describe kind of rhetoric — moral urgency, seriousness, perfect prose, as if what he was talking about right then was the most important thing that could be talked about, because of how much was at stake, while still managing to sound like a scientist."

This is a beautiful tribute to Marty.



Cade Sivertsen 20m

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Thank you for this article - I've seen several posts by professors of the "paper overnight with Claude" attempt. How can an early-career researcher stay competitive in the age of AI tools (aside from explicitly using AI tools to do all of my work)? I'm not a PhD, but I have two master's degrees and one single-authored publication, and I'm considering applying to PhD programs. I always thought that having a published paper would be a meaningful differentiator in PhD applications but now it's plausible that applicants could produce many more papers by heavily relying on AI for idea generation, analysis, and writing. I suppose applications may require more proof of technical ability to weed out those who heavily rely on AI.

What advice would you give to someone trying to build research skills or become a researcher in this environment where AI can fully automate most of the research process? The incentives to study research methodologies and coding are shrinking, but I would love to hear what an expert thinks a newcomer should do to build a career in this area.